Data communications (Lab 7): Manchester and Differential Manchester Encoding

Objective: To generate and analyze a Manchester Encoded signal using MATLAB from a given binary data sequence.

Theory

Manchester Encoding is a digital line coding technique in which each bit period is divided into two equal halves, and there is always a transition in the middle of each bit.

In Manchester encoding:

- Binary 1 → Low to High transition (−1 to +1)
- Binary 0 → High to Low transition (+1 to −1)

This means:

- Every bit contains a voltage transition at its center
- Clock synchronization becomes easier
- No DC component is produced

MATLAB Code

```
clear;
clc;

% Input binary sequence
b = [0 1 0 0 1 0];

% Number of bits
n = length(b);

% Time vector
t = 0:0.001:n;

% Convert binary to bipolar form
for i = 1:n
    if (b(i) == 0)
        b_p(i) = -1;
    else
        b_p(i) = 1;
    end
end

% Generate bipolar waveform
for j = 1:n
    bw(j*1000:(j+1)*1000) = b_p(j);
```

```
end

bw = bw(1000:end);

% Plot bipolar data
subplot(2,1,1)
plot(t,bw)
grid on
title('Input Bipolar Signal')
xlabel('Time')
ylabel('Amplitude')

% Generate Manchester encoded waveform
for i = 1:n
    if b(i) == 1
        x(i*1000:(i+0.5)*1000) = -1;
        x((i+0.5)*1000:(i+1)*1000) = 1;
    else
        x(i*1000:(i+0.5)*1000) = 1;
        x((i+0.5)*1000:(i+1)*1000) = -1;
    end
end

x = x(1000:end);

% Plot Manchester encoded signal
subplot(2,1,2)
plot(t,x)
grid on
title('Manchester Encoded Signal')
xlabel('Time')
ylabel('Amplitude')
```

<u>Code Explanation</u>

1. Initialization

```
clc;
clear;
close all;
```

- `clc` → Clears command window
- `clear` → Removes all variables

2. Define Binary Input

```
b = [1 0 1 1 1 1 0];
n = length(b);
```

- `b` contains the binary input sequence
- `n` gives the total number of bits

3. Create Time Vector

```
t = 0:0.001:n;
```

- Creates the time axis from 0 to n
- Step size 0.001 gives a smooth plot

4. Convert Binary to Bipolar Signal

```
if (b(i) == 0)
   b_p(i) = -1;
else
   b_p(i) = 1;
```

Converts:

- $0 \rightarrow -1$
- $1 \rightarrow +1$

5. Generate Bipolar Waveform

```
bw(j*1000:(j+1)*1000) = b_p(j);
```

- Each bit is expanded over 1000 samples
- Creates rectangular waveform

6. Manchester Encoding Logic

For binary **1**:

```
x(first half) = -1;
x(second half) = +1;
```

- Transition: $-1 \rightarrow +1$ (Low → High)
- For binary **0**:

```
x(first half) = +1;
x(second half) = -1;
```

Transition: $+1 \rightarrow -1$ (High → Low)

7. Plotting

Subplot 1 → Bipolar input signal

Subplot 2 → Manchester encoded signal

<u>Result:</u>

The MATLAB program produces:

1. Bipolar digital signal
2. Manchester encoded waveform

<u>Observations:</u>

- Binary 1 → Low to High transition
- Binary 0 → High to Low transition
- Every bit contains a middle transition

Conclusion: The experiment successfully demonstrated Manchester Encoding using MATLAB. The binary sequence was converted into a signal where each bit contains a mid-bit transition. Binary 1 produced a Low-to-High transition, and binary 0 produced a High-to-Low transition, verifying the principle of Manchester Encoding.

Assignment: Write a MATLAB program to generate the Differential Manchester encoded signal for a binary sequence given as input by the user.